Computational assessment of T cell plasticity in immune-related adverse events

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BACKGROUND

Immune-related adverse events (irAEs) are serious autoimmune-like side effects of immune checkpoint blockade (ICB) therapy for cancer treatment.

Understanding the mechanisms of irAEs will allow the development of irAE treatment that does not affect the anti-cancer properties of ICB therapy.

A decrease in immuno-suppressive regulatory T cells (Tregs) is associated with various autoimmune diseases.

Hypothesis:
The development of irAEs will be correlated with a decrease in immunosuppressive regulatory T cells (Tregs).

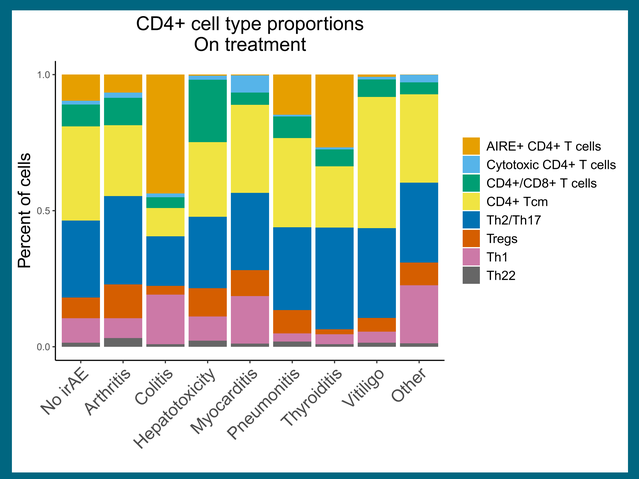
METHODS

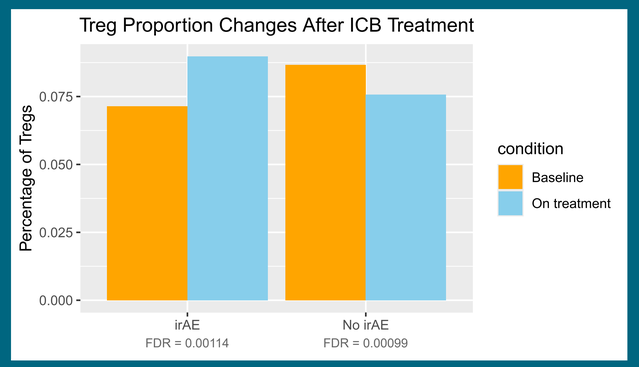
Proportion analysis of two previously published scRNA-seq datasets*

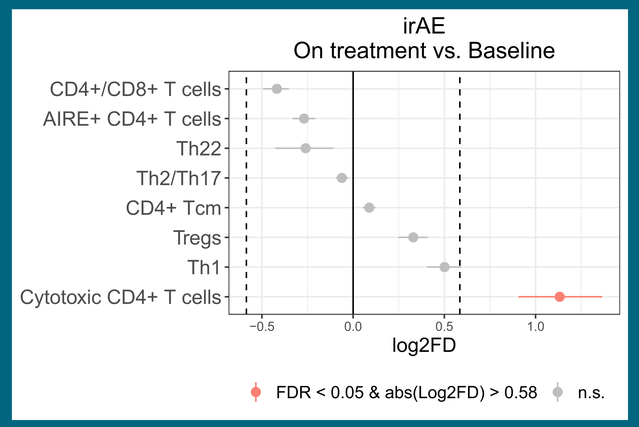
33 patients with & without irAEs before and after ICB treatment

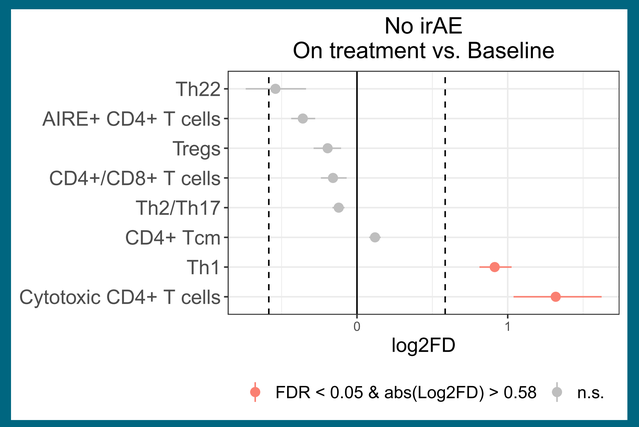
108,780 CD4+ T cells analyzed

Immune-related adverse events secondary to anti-cancer immune checkpoint blockade therapy **are associated with increased regulatory T cell proportions in circulation.**

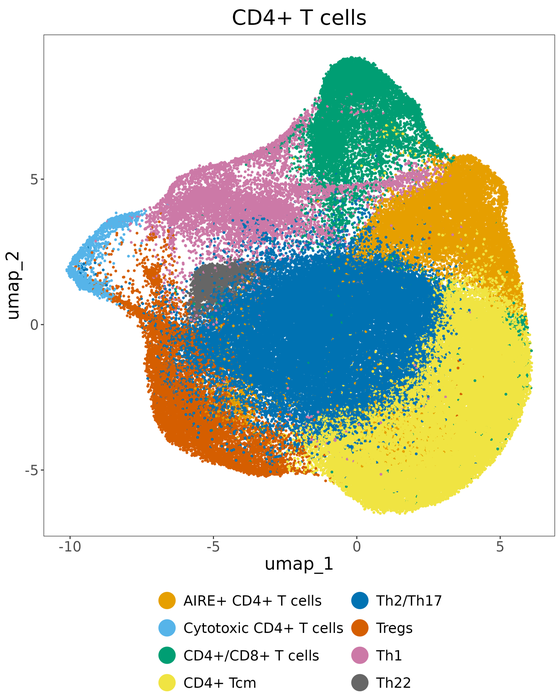








RESULTS



DISCUSSION & CONCLUSIONS

Changes in Treg populations are likely organ-specific; these datasets are underpowered for analysis of organ-specific irAEs.

We observe a small but significant increase in Tregs in the circulation of patients who develop irAEs.

Future work should examine Treg proportion changes after immune checkpoint blockade therapy in patients with and without irAEs **at the site of the irAE** rather than in circulation.

* H. Zhu et al., Circulation 146.4 (2022) doi:10.1161/CIRCULATIONAHA.121.056730
S. Bukhari et al., Cell Reports Medicine 4.1 (2023) doi:10.1016/j.xcrm.2022.100868